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Date: 11/7/2022	HILIC Data Acquisition BEH Amide Column 4 Minute Method TripleTOF 6600 Mass Spectrometry	Code No.: TTOF HILIC v1

Issued: 11/7/2022	Validity Area: Metabolomics Core
Responsible: Jeremiah Wells	Secondary:
Supersedes: HILIC Biogenic Amines analysis by UPLC-QTOF mass spectrometry	Approved:

1. Purpose: This SOP describes data acquisition for HILIC small molecule profiling using liquid chromatography/mass spectrometry (LC-MS/MS) performed on the TripleTOF 6600.

2. Equipment:

- 1) TripleTOF 6600 mass spectrometer
- 2) Agilent 1290 Infinity II UHPLC
- 3) Vortexer
- 4) Ultrasonicator
- 5) Eppendorf Centrifuge 5424
- 6) Eppendorf Single Channel Pipettes: 0.5-10 uL, 20-200 uL, 100-1000 uL, and 1-10 mL
- 7) Centrifuge for plates (Thermo Fisher Scientific)
- 8) Calibrated 8 or 12 channel pipettes and tips (Eppendorf or Thermo Fisher)
- 9) Multi-Vortexer
- 10) 96-Well Thermo Nunc plates
- 11) Plate sealer (Thermo Scientific)
- 12) 500 uL syringe

3. Chemicals and Consumables:

Product:	Manufacturer & Part Number
Acetonitrile: Optima LC/MS Grade Fisher Chemical	Fisher - A955-4, 4L
Water: Optima LC/MS grade Fisher Chemical	Fisher - W6-4, 4 L
Methanol	Fisher - A456-4
Ammonium Formate	Sigma-Aldrich - 70221-25G-F
Formic Acid: Fluka Mass Spec Grade	Fluka - F0507, 250 mL

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Acquity Premier BEH Amide 1.7um VanGuard FIT 2.1 x 50mm Column	Waters - 186009507
Acquity Premier BEH Amide VanGuard FIT Cartridge, 1.7um, 2.1 x 5 mm, 3pk	Waters - 186009510
Tuning solution	Sciex - 4456736
Taurocholic acid	Sciex - 4456736
APCI Positive Calibration Solution	Sciex - 4460131
APCI Negative Calibration Solution	Sciex - 4460134

4. Preparation of solutions

Plan to prepare 4 L of mobile phase B for every 1 L of mobile phase A.

a. Mobile Phase A (100% H₂O + 10 mM Ammonium Formate + 0.125% Formic Acid)

- i. Pre-rinse a 1 L glass bottle with pure acetonitrile and then water.
- ii. Measure exactly 1000 mL of HPLC/UHPLC MS grade H₂O in a pre-rinsed graduated cylinder and add to the 1 L glass bottle.
- iii. Weight 0.630 g of ammonium formate and add to the glass bottle. Swirl to dissolve ammonium formate.
- iv. Add 1.250 mL formic acid.
- v. Add a magnetic stir bar and mix on a stir plate for 10 minutes.
- vi. Sonicate for 10 minutes at room temperature until all the ammonium formate is dissolved.

b. Mobile Phase B (95:5 ACN:H₂O + 10 mM Ammonium Formate + 0.125% Formic Acid)

- i. Pre-rinse a 4 L glass bottle with pure acetonitrile and then water.
- ii. Measure exactly 200 mL of HPLC/UHPLC MS grade H₂O in a pre-rinsed graduated cylinder and add to the 4 L glass bottle.
- iii. Weight 2.520 g of ammonium formate and add to the glass bottle. Swirl to dissolve ammonium formate.
- iv. Add 5.00 mL formic acid
- v. Measure exactly 3800 mL of HPLC/UHPLC MS grade ACN in a pre-rinsed graduated cylinder and add to the 1 L glass bottle.
- vi. Add a magnetic stir bar and mix on a stir plate for 10 minutes. It is normal for the mixture to appear opaque/cloudy until the next step.
- vii. Sonicate for 30 minutes at room temperature until all the ammonium formate is dissolved and the solution appears clear/non-cloudy.

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c. Needle Wash Solvent (50:50 ACN:H2O)

- i. Pre-rinse a 1 L glass bottle with pure acetonitrile and then water.
- ii. Add 500 mL HPLC/UHPLC MS grade ACN to the bottle.
- iii. Add 500 mL HPLC/UHPLC MS grade H2O to the bottle.
- iv. Add a magnetic stir bar and mix on a stir plate for 5 minutes.

5. Sequence generation & organization of samples

- a. Use Carrot table acquisition generator application to create a randomized run list.
 - i. [BinView](#)
- b. Transfer the positive and negative run lists from Binview to a spreadsheet. Ensure that 1 method blank and 1 pool are injected for every 10 sample injections.
- c. For mid-sized or large studies using vials, divide the samples into batches of 60-80. Each batch will be injected in positive and negative modes overnight. Include 5 pre-run resuspension mix injections at the beginning of each mode for each batch for column equilibration.
- d. Generate vial labels which should include the following information on each vial:
 - i. Client's last name
 - ii. MX number
 - iii. WCMC sample suffix
 - iv. P#V# (Plate number and vial number. Plate number values are 1-2 and vial number values are 1-54)
- e. Print vial labels and attach to LC-MS vials.
- f. Randomize Eppendorf tubes according to the run list.

6. Preparation of samples for analysis

- a. Injection from Vials
 - i. Resuspend all samples (or the current batch of samples for larger studies) in 100 uL of resuspension solvent.
 - ii. Vortex samples for 10 seconds.
 - iii. Sonicate samples for 5 minutes at room temperature.
 - iv. Centrifuge samples for 2 minutes at 16,100 rcf.
 - v. Transfer 60 uL of each sample to a pre-labeled LC-MS vial with microinsert and cap vial.
 - vi. Transfer 30 uL of each sample to a pre-labeled eppendorf tube as a pool.

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- vii. Briefly vortex the pool and then transfer 60 uL to pre-labeled LC-MS vials, then cap vials.
- viii. Transfer LC-MS vials from blue trays to black instrument racks. Vials should be arranged going from top to bottom of each column rather than from left to right in each row.
- b. Injection from Plates
 - i. Resuspend each well with 50 uL of resuspension solvent using multi-channel pipettes.
 - ii. Immediately reseal plate using Thermo Scientific plate sealer set to 165°F for 4 seconds.
 - iii. Gently vortex for 10 seconds using plate vortexer.
 - iv. Centrifuge plate at 1540 g for 5 minutes.
 - v. The TripleTOF instrument requires Nunc plates. If resuspension plates are not Nunc, transfer 40 uL from each well to a Nunc plate using a multi-channel pipette and then seal on plate sealer.

7. Preparation of instrument

- a. Ensure that the rough pump oil has been changed within the past year. Oil changes are tracked on YouTrack.
- b. Ensure that the calibration bottles on top of the mass spectrometer are at least half full, and refill as needed with APCI Positive Calibration Solution and APCI Negative Calibration Solution (which are stored in the fridge). Expired calibration solutions are acceptable since the analytes are very stable.
- c. Ensure that the TripleTOF has been tuned in both positive and negative modes within the past month via instrument optimization. Complete the following procedure if instrument optimization is needed.
 - i. Open Analyst software if not already open.
 - ii. Deactivate any active hardware configurations.
 - iii. Activate the 6600 hardware configuration.
 - iv. Flush a 500 uL syringe 5x with methanol, along with the corresponding infusion PEEK tubing.
 - v. For positive mode, aspirate 500 uL of Sciex Tuning Solution into the syringe. For negative mode, aspirate 500 uL of Sciex Taurocholic acid into the syringe. These solutions are stored in the fridge. Do not use expired Tuning Solution or Taurocholic acid for this procedure.

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- vi. Install the syringe in the built-in syringe pump on the front of the TripleTOF and then plumb the PEEK tubing to attach to the inlet of the TripleTOF source.
- vii. Double-click "Manual Tuning" in Analyst.
- viii. Change the project folder to API Instrument.
- ix. Open Postoftuning.dam (for positive mode) or Negtoftuning.dam (for negative mode)
- x. Ensure that the tuning method conforms to the following parameters:
 1. GS1: 20
 2. GS2: 20
 3. Curtain Gas: 20
 4. Temperature: 150
 5. IonSpray voltage: 5500 (positive mode) or -4500 (negative mode)
 6. DP: 80 (positive mode) or -80 (negative mode)
 7. CE: 10 (positive mode) or -10 (negative mode)
 8. Syringe diameter: 3.260
 9. Flow rate: 7 uL/min
- xi. Click "Start Syringe" and the syringe pump will begin pumping
- xii. Click "Start" and the MS will begin acquiring data.
- xiii. Observe the TIC and wait for the signal to stabilize. It typically will take around 5-10 minutes for the TIC to stabilize. Before that time, the signal will often fluctuate significantly as bubbles are flushed out of the line.
- xiv. Once the signal is stable, stop the acquisition (but leave the syringe pump running). Click "Instrument Optimization" to begin the tuning process.
- xv. Select mode.
- xvi. Select the appropriate tuning solution: Csl_ALILTLVS_Reference (positive mode) or Taurocholic acid (negative mode).
- xvii. "Verify Performance Only" should remain unchecked. The checkboxes under TOF MS, Product Ion - High Resolution, Product Ion - High Sensitivity, and Detector should all remain checked.
- xviii. Click Next.
- xix. Ensure that the following parameters are specified:
 1. GS1: 20
 2. GS2: 20
 3. IonSpray voltage: 5500 (positive mode) or -4500 (negative mode)
 4. Syringe diameter: 3.260

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5. Flow rate: 7 uL/min
 - xx. Click "GO!" and instrument optimization will begin. Optimization will take around 30-45 minutes per mode.
 - xxi. A report will be displayed following completion of instrument optimization. The top of the report displays instrument performance metrics prior to optimization. The bottom of the report displays performance metrics after optimization. Mass accuracy values are typically < 5 ppm after instrument optimization.
- d. Ensure LC is working properly
- i. When installing a brand new column:
 1. Purge the pumping system of any old buffers and connect the inlet of the column to the injector outlet. Attach the outlet line, but allow flow to go into a beaker instead of to the ion source of the mass spectrometer.
 2. Flush column with 100% Mobile Phase B with a pump flow rate at 0.8mL/min over 10 minutes.
 3. Switch to 70% mobile phase A and flush the column for 10 minutes.
 4. When the mobile phase is flowing freely from the column outlet, stop the flow and attach the column outlet to the ion source of the mass spectrometer.
 5. Switch back to 100% Mobile Phase B and monitor the back pressure until a steady value is achieved. Expected values are around 3000-3500 psi
 - ii. When completely switching mobile phases:
 1. Disable any active hardware profiles in Analyst.
 2. Open Agilent Lab Advisor.
 3. Click "Connect" followed by "Instrument Control".
 4. Click the arrow underneath 1290 Bin Pump to directly control the pump.
 5. Set channels appropriately.
 6. Set Composition to 100% A and then click Prime to "On". The pump will then purge solvent for about 3 minutes. Repeat once.
 7. Set Composition to 100% B and then click Prime to "On". The pump will then purge solvent for about 3 minutes. Repeat once.
 8. Click "System Overview" followed by "Disconnect".
 9. Reactive the appropriate hardware profile in Analyst.

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- iii. Ensure that the needle wash bottle is at least 50% full. If it runs dry, a syringe will be needed to prime the line.
- iv. Click “Acquire” in Analyst. Open the CORE_HILIC project folder.
- v. Equilibrate the column as follows:
 1. Open Agilent_manual.dam method and change flow rate to 400 uL/min at 100% B. Save the method.
 2. Click the equilibration button and allow pressure to stabilize.
 3. Change the flow rate to 800 uL/min at 100% B and save the method.
 4. Click the equilibration button and allow pressure to stabilize.
Approximate pressure variation should be < 25 psi after equilibration. If pressure oscillation is observed, a bubble has likely entered the system. Complete the following procedure to purge the bubble:
 - a. Turn off the LC flow.
 - b. Connect a union (aka zero dead volume fitting) in place of the column.
 - c. Disconnect the PEEK tubing from the source and transfer to a beaker so that LC flow will be directed to the beaker instead of the mass spectrometer.
 - d. Change the flow rate to 4000 uL/min at 50% B and save the method.
 - e. Click the equilibration button and allow the LC to flow for about 5 minutes.
 - f. Stop LC flow.
 - g. Follow steps 1-4 above and check to ensure that pressure is now stable.
- vi. High or incrementally increasing pressure may indicate the need for changing the needle seat. If this is required, follow procedure in given document:
 1. [Exchange Needle Seat](#)

8. Run sequence

- a. Ensure that the “View Queue” icon has been clicked to display the sample queue. Other functions in Analyst will not be available unless this icon has first been clicked.
- b. Click “Build Acquisition Batch” to generate a new sequence.
- c. Click “Add Set”. Click “Add Samples” and enter the number of injections that will be included in your batch.

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- d. Ensure that the plate code is *54VialPlate* for all samples when using vials or *96Nunc* when using Nunc 96-well plates.
- e. Injection volume can range between 1 uL and 5 uL. Plasma and serum are always injected at 5 uL for both modes. For other matrices, perform a test on the pool in positive mode to determine optimal injection volume. Using injections of plasma or serum from previous studies as a guide for expected TIC visual appearance and intensity, decide on an optimal injection volume for the new matrix. The optimal volume determined for positive mode can also be used for negative mode.
- f. Select the appropriate method for the batch:
Positive mode: HILIC_DDA5_POS_A2B2_M1_ADVR_OFF-shortcolumn.dam
Negative mode: HILIC_DDA5_NEG_A2B2_M1_ADVR_OFF-shortcolumn.dam
- g. Copy and paste the data file names generated in Binview to the Sample Name and Data File columns in the newly created batch.
- h. Copy and paste the vial and plate positions into the newly created batch. Enter the appropriate injection volumes.
- i. Click the "Calibrate" tab. Ensure that "Auto Calibration" is checked. Calibration should be performed once for each mode for every overnight batch, so enter an arbitrarily large number (250) into the "Calibrate Every X Samples" box.
- j. Click the "Submit" tab. Click the "Submit" button to start sample analysis.
- k. Check the sample queue to ensure that the batch has been properly submitted and includes a calibration.
- l. Wait to verify that the calibration completes successfully (which is indicated by a green check mark) before moving on to other activities.
- m. After submitting positive mode, repeat steps f-k for negative mode.
- n. It is good practice to check that solvent is observed in the wash port after the first injection to ensure the needle wash system is working properly.
- o. Allow positive and negative mode injections to run overnight and then perform quality control the following day before moving on to another batch/study

9. Perform quality control

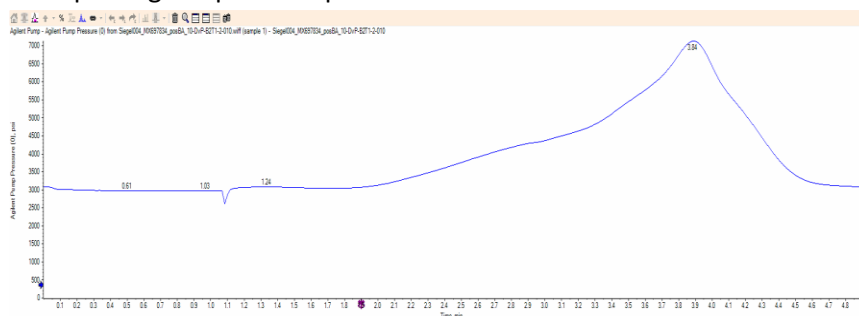
- a. Use the mz/rt correction sheets in the following location to quality control the internal standards peak intensity, shape, retention time, and mass accuracy:
 - i. metabolomics\Data Processing\mzRT Libraries\HILIC 5 minute method
- b. Copy the positive mode files to an appropriately labeled folder, and then the negative mode files to another folder.

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c. Use Sciex PeakView to QC data

- i. Open all sample files from a given batch for one mode.
- ii. Check the TICs of the overlaid samples to ensure there were no bad injections. A bad injection would be indicated by a flatline TIC. Re-inject vials as needed (again with 5 equilibration injections of resuspension solution).
- iii. Open all pool files from a given batch for one mode. Check for TIC consistency.
- iv. Open all blanks from a given batch for one mode. Extract ion chromatograms of an early-eluting iSTD (CUDA), a middle-eluting iSTD (D3-Creatinine), and a late-eluting iSTD (D7-Arginine).
- v. Check that iSTD peak heights look reasonably consistent, and ensure retention times of iSTDs are within a reasonable time shift from the mz/rt (+/- 0.3 minutes typically) and do not drift considerably across the study.
- vi. Look at the pressure profile of the blanks by clicking Show > UV/DAD/ADC Channel. If the pressure profile is inconsistent and fluctuates, check if there is an issue with the LC by inspecting for blockages, changing the needle seat, or running clean mobile phase through the system with a union at a high flow rate for several minutes.
 1. Variable pressure does not necessarily indicate an injection is bad. Use standards and TIC for this. The pressure profile is more to determine if LC maintenance needs to be done.

vii. Example of good pressure profile:



viii. Repeat i-vii for the other mode.

10. After data acquisition & QC

- a. Create a folder labeled with the MiniX id of the study under the correct year in the Eclipse Core drive.

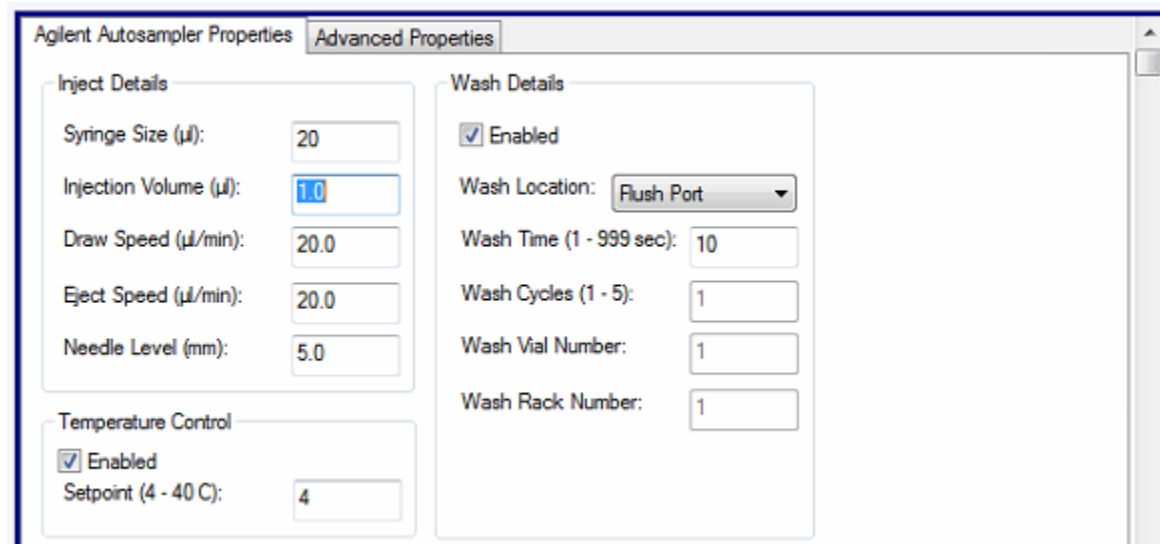
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- b. Copy the folder containing the data files on the local PC to the newly created folder on the Eclipse Core drive.

11. Data acquisition methods

	Total Time (min)	Flow Rate (μ l/min)	A (%)	B (%)
0	0.00	800	0.0	100.0
1	0.50	800	0.0	100.0
2	1.95	800	30.0	70.0
3	2.55	800	70.0	30.0
4	3.15	800	0.0	100.0
5	3.80	800	0.0	100.0

a.



Agilent Autosampler Properties - Advanced Properties

Inject Details

Syringe Size (μ l): 20

Injection Volume (μ l): 1.0

Draw Speed (μ l/min): 20.0

Eject Speed (μ l/min): 20.0

Needle Level (mm): 5.0

Temperature Control

☒ Enabled

Setpoint (4 - 40 C): 4

Wash Details

☒ Enabled

Wash Location: Flush Port

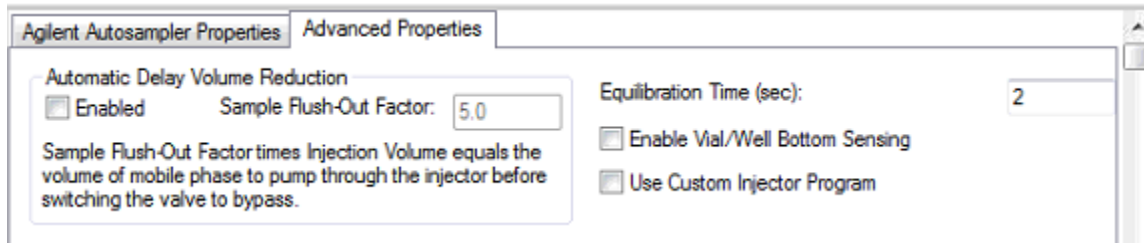
Wash Time (1 - 999 sec): 10

Wash Cycles (1 - 5): 1

Wash Vial Number: 1

Wash Rack Number: 1

b.



Agilent Autosampler Properties - Advanced Properties

Automatic Delay Volume Reduction

☒ Enabled Sample Flush-Out Factor: 5.0

Sample Flush-Out Factor times Injection Volume equals the volume of mobile phase to pump through the injector before switching the valve to bypass.

Equilibration Time (sec): 2

☐ Enable Vial/Well Bottom Sensing

☐ Use Custom Injector Program

c.

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d.

Agilent Column Oven Properties

Temperature

Left Temperature (°C): 45.0

Right Temperature (°C): 45.0

Temperature Tolerance +/- (°C): 1.0

Start Acquisition Tolerance +/- (°C): 5.0

Column Switching Valve

☐ Time Table Used

Position for first sample in the batch: Left

☒ Use same position for all samples in the batch

☐ Alternate position for each sample in the batch

	Total Time(min)	Valve Position
0		
1		
2		
3		
4		
5		
6		

Add Entry Remove Entry

e.

MS Advanced MS Switch Criteria Include/Exclude IDA Advanced

Experiment: 1 ☒ IDA Experiment Create IDA Exp Create SWATH™ Exp

Scan type: TOF MS

Accumulation time: 0.100007 (secs)

TOF Masses (Da)

Min: 50 Max: 1500

Polarity

☒ Positive ☐ Negative

Edit Parameters

Period

Duration: 3.802 (mins) Cycles: 912 Delay Time: 0 (secs)

Cycle time: 0.2501 (secs) Period: 1

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MS

Advanced MS

Switch Criteria

Include/Exclude

IDA Advanced

Experiment:

2

IDA Experiment

Create IDA Exp

Create SWATH™ Exp

Scan type:

Product Ion

Product Of:

IDA

Accumulation time :

0.020024

(secs)

Polarity

Positive

Negative

Edit Parameters

TOF Masses (Da)

Min:

40

Max:

1000

High Resolution

High Sensitivity

Enhance Mass

	Mass (Da)	Enhance
1		

Period

Duration:

3.802

(mins)

Cycles:

912

Delay Time:

0

(secs)

Cycle time:

0.2501

(secs)

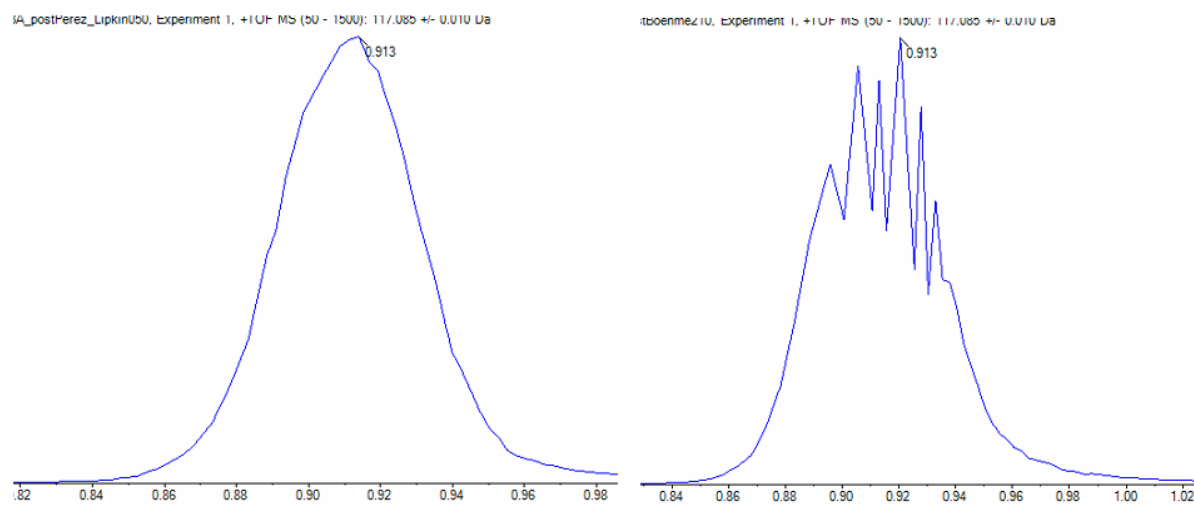
Period:

1

f.

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- a. The most commonly observed problems include calibration failure (typically negative mode) and deterioration of smooth peak shape into jagged peak shape.
- b. Deterioration of peak shape into a jagged appearance occurs when the QJet ion guide becomes dirty and is due for a cleaning. D3-Creatinine is a particularly good diagnostic ion for monitoring this problem. An example of good iSTD peak shape is shown below on the left, with the deteriorated jagged peak shape shown on the right.



- c. To clean the QJet ion guide, follow the procedure found in the following manual (also available via google):
 - i. `metabolomics\Sciex\instrument-front-end-cleaning-customer-en.pdf`
- d. If the automated calibration fails (indicated by a red X in place of a green check mark on the sample queue), check to ensure that enough calibrant is present and refill if necessary and then repeat the calibration. If calibration still fails, use the following procedure to perform a manual calibration:
 - i. `metabolomics\Sciex\Manual Calibration Using the Tune Method Editor – Sciex 6600 TTOF_2.docx`
- e. In order to avoid cross-contamination and artifact formation, disposable consumables are used (Eppendorf plastic tubes, plastic pipette tips).
- f. If experiencing TripleTOF issues that seem software related, try the following:
 - i. Restart the acquisition software
 - ii. Restart the computer and re-open acquisition software
 - iii. If still experiencing issues, power cycle the TripleTOF. Follow documentation provided by Sciex:

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1. metabolomics\Sciex\6600-system-user-guide-en.pdf

13. Disposal of waste

- a. Chemicals are disposed into appropriate containers in the lab in secondary confinement before monthly disposal collection. Glass vials and other consumables are collected in the plastic bags and stored in the lab before monthly disposal.